

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0875

Metaline Falls 100 kV
April 23, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0875. The Interconnection Customer proposes a 76.2 MW solar photovoltaic facility interconnecting at the Metaline Falls 100 kV substation in the Ohio area. The request was received on October 10, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 316 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,384,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Minidoka / Bannock Crk 100 KV	Gilcrest / Metaline Falls 100 KV	124.4	138	100.0
Metaline Falls / Bannock Crk 100 KV	Metaline Falls / Gilcrest 100 KV	122.2	140	100.0
Woburn / Gilcrest 100 KV	Fort Bragg / Wells 230 KV	120.1	223	100.0
Gilcrest / Metaline Falls 100 KV	Wells / Ft Bragg 230 KV L	118.8	232	100.0
Tontitown / Wilbraham 100 KV line	Ravenna / Hylebos Substation 500 KV line	114.5	88	14.5
Wilbrham / Yorba Linda 100 KV line	Wells / Fort Bragg 230 KV line	106.8	95	58.4
Ft Bragg / Minidoka 100 KV	Metaline Falls / Gilcrest 100 KV	103.9	155	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Minidoka / Bannock Crk 100 KV	Rebuild line with higher-capacity conductor	\$4,988,000	42% (\$2,108,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Metaline Falls / Bannock Crk 100 KV	Rebuild line with higher-capacity conductor	\$4,113,000	42% (\$1,738,000)
Woburn / Gilcrest 100 KV	Rebuild line with higher-capacity conductor	\$9,181,000	42% (\$3,880,000)
Metaline Falls / Gilcrest 100 KV	Rebuild line with higher-capacity conductor	\$11,989,000	42% (\$5,067,000)
Tontitown / Wilbraham 100 KV L	Rebuild line with higher-capacity conductor	\$10,123,000	33% (\$3,302,000)
Wilbrham / Yorba Linda 100 KV line	Reconductor line	\$1,753,000	42% (\$741,000)
Ft Bragg / Minidoka 100 KV	Reconductor line	\$1,298,000	42% (\$548,000)
	Total allocated to project		\$17,384,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.